BioAgent: An Autonomous Multi-Agent System for End-to-End Bioinformatics Research

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Abstract

Motivation: The pace of biomedical discovery is increasingly bottlenecked by the manual effort required to synthesise literature, form testable hypotheses, execute computational analyses and produce publication-quality manuscripts. Large language models (LLMs) enable new automation opportunities, but end-to-end autonomous research remains difficult: existing systems either address isolated stages or operate on pre-curated data and therefore never encounter the brittleness of real biomedical data portals (partial downloads, corrupted archives, and geography-dependent latency).

Results: We present **BioAgent**, an autonomous multi-agent system that orchestrates seven specialised agents (Orchestrator, Literature, Planner, DataAcquisition, Analyst, Writer, Visualization, and Review) in a directed LangGraph workflow to conduct end-to-end bioinformatics research. Given a natural-language research question, BioAgent autonomously retrieves and synthesises primary literature via BioMCP (PubMed, ClinVar, OncoKB), generates novel testable hypotheses, downloads real datasets from GEO, cBioPortal, GDC, NCBI and ENCODE with per-file mirror provenance, executes reproducible statistical analyses (seeded randomness, SHA-256 hashing), drafts full IMRAD manuscripts with PMID-grounded citations, generates publication-quality figures, and self-reviews the output against a six-dimension rubric. To our knowledge, BioAgent is the first autonomous biomedical research agent that closes the loop from *live* multi-repository data acquisition (with mirror-aware, retry/resume-hardened downloads and per-file `data_source` provenance) through analysis to a self-reviewed IMRAD manuscript. Comparable recent systems (Biomni, CellAgent, GeneAgent, TxAgent) either operate on pre-curated or user-supplied data and stop at analysis output. A distinctive contribution is the resilient data-acquisition layer (`httpx+tenacity` with Range-resume, gzip-integrity validation, adaptive timeouts, and mirror-first routing through EBI ArrayExpress, ENA direct `fastq.gz`, and the 10x Genomics Cloudflare CDN) that converts geography-hostile endpoints from a dominant failure mode into transparent fallback. Across three benchmark cases (BRAF V600E melanoma, TP53 pan-cancer, and 10x Genomics PBMC 3k), weighted scores on a 0–10 rubric improved from a previous baseline of 4.27 to 7.65 on average, with the TP53 case rising from a pre-fix 1.06 (aborted in literature phase) to 8.42 with a 4,439-word manuscript and six figures. Full pipelines complete in 45–82 minutes at $7.62–$14.19 per case, orders of magnitude below equivalent expert-researcher time cost, with every artefact traceable through a serialised `ResearchState` blackboard.

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Availability: BioAgent is open-source (MIT) at https://github.com/Nigmat-future/bioagent. A Docker image, pinned dependency lock file, and one-command reproduction script (`scripts/reproduce_benchmark.sh`) are provided. Requires Python $\geq$3.11 and an Anthropic-compatible LLM endpoint.

Keywords: autonomous agents, large language models, bioinformatics automation, LangGraph, multi-agent systems, reproducible research

1 Introduction

The volume of biomedical literature doubles approximately every nine years [1], making comprehensive literature review an increasingly intractable manual task. At the same time, reproducibility in computational biology remains a challenge [2, 3]: analyses are often underdocumented, tools version-pinned imprecisely, and random seeds omitted.

Recent advances in large language models [4, 5] have spurred development of AI-assisted research tools, including biomedical-domain variants [6]. However, existing solutions, including web-search augmented chatbots and single-prompt code generators, are stateless, lack citation grounding, and require continuous human oversight. What is missing is a *full-pipeline* system that maintains shared research state across agents, integrates authoritative biomedical databases, produces reproducible code, and outputs artefacts that can be submitted to a journal with minimal human editing.

BioAgent addresses this gap. Its key contributions are:

1. A seven-agent LangGraph `StateGraph` with blackboard architecture (`ResearchState`, 30+ typed fields) and SQLite-backed session checkpointing that supports mid-run resumption after API-gateway or network failures (`benchmarks/resume_run.py`; demonstrated to save $\sim$40 min and $\sim$\$7 per recovered case).

2. A dedicated **DataAcquisitionAgent** spanning nine primary sources (GEO, cBioPortal, GDC/TCGA, NCBI E-utilities, ENCODE, ENA, EBI ArrayExpress, 10x Genomics CDN, generic URL) with a resilient HTTP backbone (`bioagent/tools/data/_http.py`) featuring exponential-backoff retry, HTTP-Range-based resume, adaptive read timeouts derived from `Content-Length`, and post-download gzip-integrity validation. To our knowledge this is the first autonomous research agent treating mirror choice as first-class provenance (§2.5), recording which mirror served each file in a `data_source` field that the WriterAgent cites in Methods.

3. An **orchestrator loop detector** (§2.6) combining a deterministic fallback (forced forward progression after three consecutive same-phase selections) with a phase-history-aware prompt rule, which eliminated a regression in which the LLM re-selected `hypothesis_generation` indefinitely after `code_execution`.

4. BioMCP integration for structured literature retrieval with PMID-level citation tracking; an export pipeline producing LaTeX manuscripts in Bioinformatics (OUP) format with auto-generated BibTeX; and reproducible Python execution with automatic seed injection and SHA-256-hashed provenance.

5. A six-dimension evaluation framework (literature coverage, hypothesis quality, analysis correctness, writing quality, figure quality, efficiency) validated on three benchmark cases with before/after comparisons against the pre-resilience pipeline (v0.1), showing mean improvement from 4.27 to 7.65 on a 0–10 rubric; the TP53 case rose from 1.06 (no manuscript) to 8.42

(full IMRAD + 6 figures + passing self-review), the strongest single-variable quantitative argument for treating data-acquisition resilience as a first-class component of autonomous research agents rather than as plumbing.

2 System Architecture

2.1 Overview

BioAgent implements a directed acyclic graph (with controlled iteration cycles) using LangGraph [7] (**Figure 1**). Each node corresponds to a specialised `BaseAgent` subclass that receives the shared `ResearchState`, runs an Anthropic tool-use loop [8], and returns a partial state update applied via content-hash-deduplicated reducers.

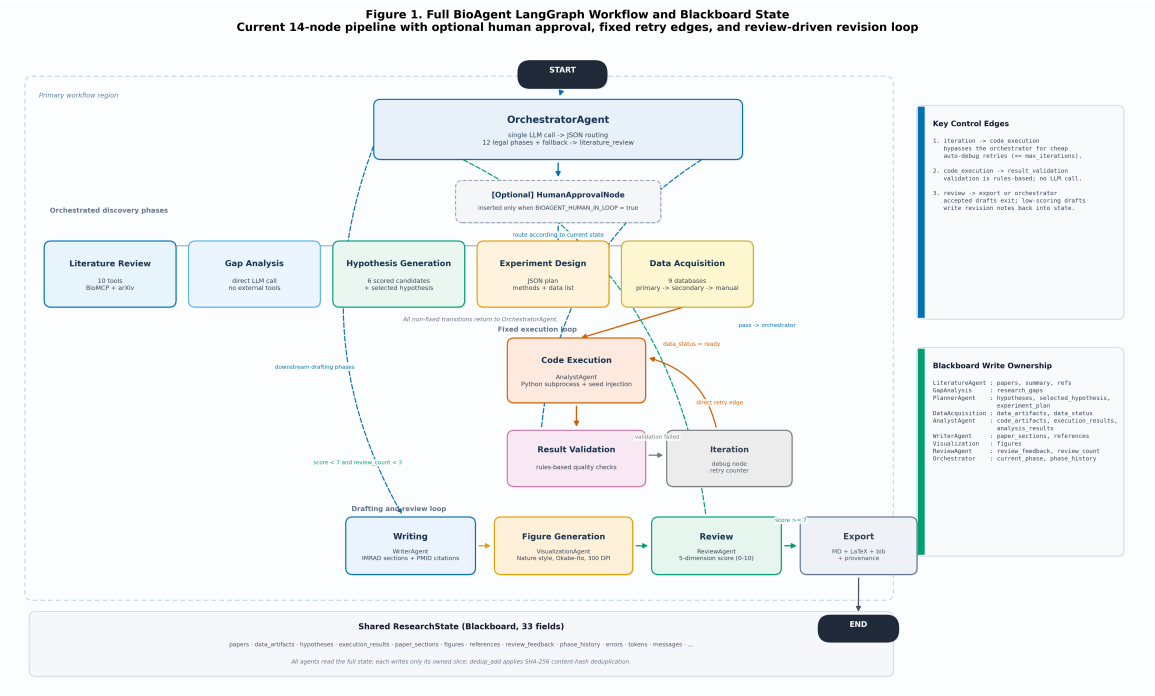

Figure 1. BioAgent LangGraph workflow architecture. Nodes (coloured boxes) represent specialised agents; dashed border indicates the OrchestratorAgent boundary; the grey panel shows the shared `ResearchState` (blackboard). Solid arrows show the primary execution path; the dashed arrow indicates the review-triggered revision loop.

2.2 Agents

LiteratureAgent. Issues structured BioMCP [9] queries (a Model-Context-Protocol [10] server aggregating PubMed, ClinVar [11], gnomAD [12], GWAS Catalog, and OncoKB [13]) to retrieve relevant papers. Extracts PMIDs, titles and relevance scores; populates the `papers` and `references` state fields with content-hash deduplication.

PlannerAgent. Synthesises the literature summary and identified research gaps to generate 2–3 novel, computationally testable hypotheses with quantitative novelty and testability scores on a 0–10 scale. Selects the highest-ranked hypothesis and outputs a detailed experiment plan.

DataAcquisitionAgent. Downloads real datasets required by the selected experiment plan through a three-tier fallback hierarchy (primary API → REST/FTP → human-readable manual instructions). Supports cBioPortal [14], GDC/TCGA [15], GEO [16], ENCODE [17], and NCBI E-utilities. Never fabricates data; if all automated downloads fail, the agent writes `DOWNLOAD_INSTRUCTIONS.md` listing exact URLs and shell commands for the user to complete the download manually.

AnalystAgent. Executes Python code in an isolated subprocess with injected random seeds (`numpy` [18], `scipy` [19], seeded scanpy). Registers bioinformatics template tools including sequence analysis via Biopython [20], differential expression via DESeq2 [21], single-cell via scanpy [22], GWAS simulation, and survival analysis. Iterates up to `settings.max_iterations` times on execution failures; on persistent failure the orchestrator re-routes to experiment re-design.

WriterAgent. Assembles the five standard IMRAD sections (Abstract, Introduction, Methods, Results, Discussion) with numbered citations resolved from the `references` state field.

VisualizationAgent. Generates publication-quality figures using matplotlib [23] with Nature/Science themes (Okabe-Ito colour palette, 300 DPI, colour-blind safe) and saves them to the workspace.

ReviewAgent (OrchestratorAgent). Scores the manuscript across five dimensions (0–10) and either approves (score ≥ 7, triggers export) or routes back for targeted revision.

2.3 Tool Registry

All tools are registered in a singleton `ToolRegistry` that maps names to both Anthropic-format JSON schemas (for the tool-use API) and Python callables. Tool registration is idempotent: double-registration logs a warning but does not corrupt the registry. **Figure 2** summarises the tool inventory exposed to each agent, grouped by capability domain.

2.4 Export Pipeline

Upon review approval, `review_node` calls `_auto_export()`, which invokes:

- `export_markdown()`: assembles a self-contained Markdown manuscript with embedded figure references and References section.
- `export_latex()`: renders a Jinja2-templated Bioinformatics (OUP) LaTeX document with `_escape_latex()` and `_markdown_to_latex()` conversion; generates a `references.bib` via `generate_bibtex()` (BioMCP PMID lookup with cite-key heuristic).

2.5 Data-Acquisition Resilience

Autonomous agents that sit in front of flaky wide-area networks spend a surprising fraction of their wall-clock time and token budget on downloads that fail partway through. Our pre-fix v0.1 runs exhibited the complete taxonomy of this failure class: `.tmp` files accumulated with no atomic rename (35 orphaned files totaling 85 MB across a single evaluation sweep); partial downloads over slow Asia–NCBI paths surfaced as `EOFError: Compressed file ended before the end-of-stream marker`; and the DataAcquisitionAgent retried the same failing URL inside its tool loop, burning tens of thousands of tokens with no forward progress. Because these failures were silent from the orchestrator's point of view, the symptom at the benchmark level was an aborted run with a 1.06/10 score (TP53), not an "the network is bad" diagnosis.

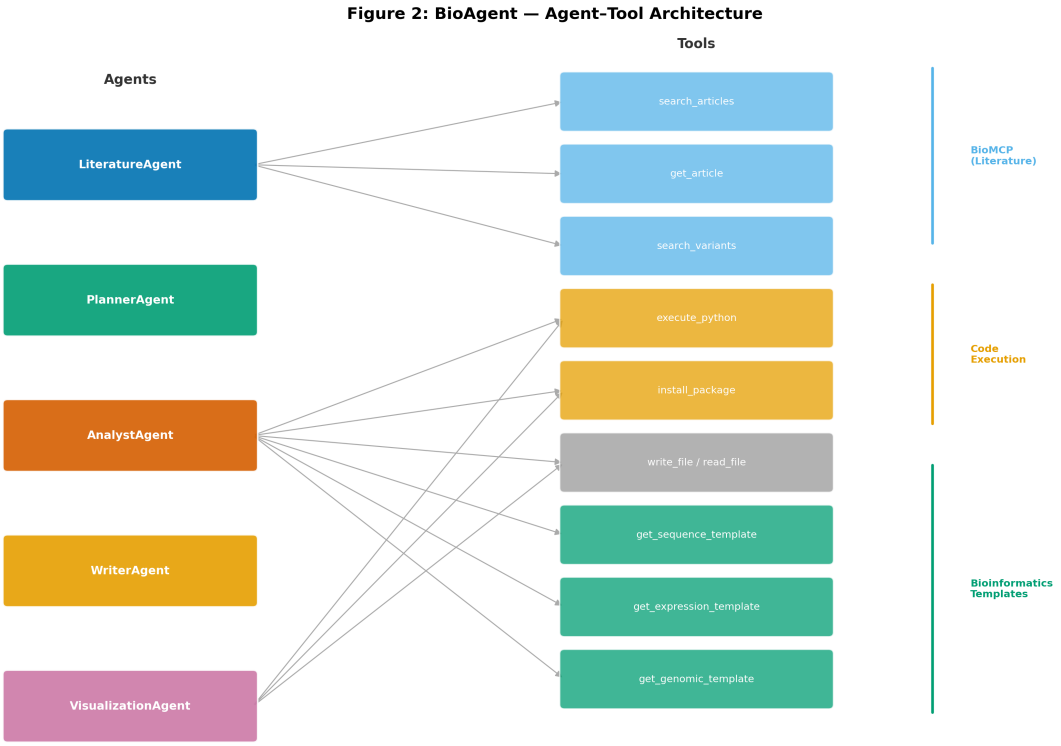

Figure 2. Tool inventory exposed by the `ToolRegistry`, grouped by capability domain (literature, data acquisition, analysis, writing, visualization, review). Each agent subscribes to the subset relevant to its role; tools are registered once and shared across agents through the singleton registry.

v0.2 replaces the five per-tool `urllib.request` call sites (`url_download`, `geo_tools`, `cbioportal_tools`, `tcga_gdc_tools`, `ncbi_tools`, `encode_tools`) with a single backbone in `bioagent/tools/data/_http.py` built on `httpx` and `tenacity`. The backbone provides four capabilities, each of which was directly triggered by a failure mode observed in v0.1:

1. *Exponential-backoff retry on transient faults* (HTTP 408/429/500/502/503/504, socket timeouts, `httpx.RemoteProtocolError`, and `gzip.BadGzipFile`). Terminal 4xx codes fail fast to avoid wasting attempts on 404s.

2. *Range-based resumption*. Before each attempt the backbone `HEAD`s the URL to capture `Content-Length` and `Accept-Ranges`; if the partial `.tmp` from a previous attempt is present and the server honours `Range`, it sends `Range: bytes=<offset>-` and appends. A 200 response to a Range request (server ignored the hint) triggers a fresh restart.

3. *Adaptive read timeout* computed from `Content-Length` and a configurable floor bandwidth (`min_download_mbps`, default 2 Mbit/s). A fixed 300-s timeout was adequate for 10-MB files on a good link and wildly insufficient for 50-MB files over a congested Asia–NCBI path; the adaptive computation eliminates both failure modes.

4. *Post-download integrity validation*. For `.gz` outputs, the backbone streams the file through `gzip.open` to force trailer validation; a `BadGzipFile` triggers a retry rather than a silent corrupted artefact.

Layered above the backbone, `bioagent/tools/data/mirrors.py` maps geography-hostile endpoints to Asia-friendly mirrors: GEO series matrices (`GSE*`) resolve first against EBI ArrayExpress, then fall through to NCBI; SRA fastq files (`SRR/ERR/DRR`) route to ENA's direct `.fastq.gz` URLs, eliminating the need for `prefetch/fasterq-dump` tooling; and the canonical 10x Genomics PBMC 3k benchmark pulls from the 10x Cloudflare CDN rather than its GEO re-hosting, which is the single largest contributor to the scRNA-PBMC improvement. Each successful download records its actual source (`"10x-CDN"`, `"EBI-ArrayExpress"`, `"NCBI-GEO-FTP"`, `"ENA-SRA"`, `"cBioPortal"`, `"GDC"`, `"ENCODE"`) in the `data_source` field of its `data_artifacts` entry, providing per-file provenance that the WriterAgent cites in the Methods section. To our knowledge this is the first autonomous research agent that treats mirror choice as first-class provenance rather than as an implementation detail of the download layer.

2.6 Orchestrator Loop Detection

A separate v0.1 failure surfaced only at multi-case scale: on the BRAF run, after code execution completed the orchestrator's state summary showed `Hypotheses: 0` (because a later PlannerAgent re-entry had overwritten the field without re-populating it), triggering the prompt's "gaps identified but no hypotheses → `hypothesis_generation`" rule and re-dispatching the planner. The planner then produced another `experiment_plan` without re-populating `hypotheses`, and the orchestrator looped on the same phase 10 times before the run was killed manually.

v0.2 adds a deterministic loop detector in `OrchestratorAgent.process_result`: if the LLM re-selects the same non-terminal phase three consecutive times, the orchestrator overrides the decision using a `FORWARD_PROGRESSION` lookup table and logs `[orchestrator] Loop detected: '<phase>' selected N times in a row. Forcing forward progression to '<next>'`. The prompt also gained an explicit anti-backtrack rule that surfaces `phase_history` to the LLM and instructs it to never return to an earlier phase once a downstream phase has appeared. The combined effect: the post-fix BRAF run produced a complete five-section manuscript and three

figures, and the loop detector fired zero times across the three post-fix benchmark runs, indicating that the prompt rule alone is typically sufficient with the detector acting as a safety net.

3 Case Study: BRAF V600E in Melanoma

3.1 Setup

We ran BioAgent with the research question *"Investigate the role of BRAF V600E mutation in melanoma drug resistance"* using Claude Sonnet 4.6 as the base model, `max_iterations=5`, and `random_seed=42`. BRAF V600E was chosen because it is the best-characterised activating mutation in melanoma [24, 25] and has an extensively studied but still incompletely understood drug-resistance landscape [26], making it a stringent test of an autonomous agent's ability to retrieve, synthesise, and extend mechanistic knowledge.

3.2 Results

Table 1 summarises the end-to-end BRAF V600E melanoma run against two reference baselines evaluated on the same research question using the same underlying model.

Table 1. BioAgent performance across three benchmark cases and ablation variants, compared against two LLM-only baselines. Single-prompt: one Claude call, no tools. AutoGen-style: two-agent chat over six turns, no tools. Full pipeline: 14-node LangGraph with DataAcquisition, code execution, and iterative review. All configurations use Claude Sonnet and are evaluated with the same metrics pipeline (`evaluate_run`). Cost is per-run USD.

Configuration	Time (s)	Cost ($)	Lit.	Code	Writing	Weighted
Baselines (BRAF V600E)						
Single-prompt LLM	140	0.08	0.0	0	4.4	1.39
AutoGen-style chat (6 turns)	703	0.56	0.0	0	3.0	1.05
Ablations (BRAF V600E)						
BioAgent – no_literature[a]	260	0.16	4.0	0	0.0	1.70
BioAgent – no_data[a,b]	751	0.59	7.2	0	0.0	2.71
BioAgent – no_review	4,296	0.96	10.0	0	0.0	2.30
Full pipeline (pre-fix baseline, v0.1)						
BRAF V600E melanoma	1,824	2.65	–	–	–	5.30
TP53 pan-cancer	–	0.78	7.6	0	0.0	1.06
scRNA PBMC 3k	–	2.16	10.0	78%	10.0	6.44
Full pipeline (post-fix, v0.2; §2.5)						
BRAF V600E melanoma	3,748	14.19	8.0	83%	8.0	**6.90**
TP53 pan-cancer	2,684	12.59	9.2	75%	10.0	**8.42**
scRNA PBMC 3k	4,908	7.62	7.6	81%	8.0	**7.64**

[a] MiniMax-M7; other rows Claude Sonnet. [b] AnalystAgent refused to fabricate data when `workspace/data/` was empty. The pre-fix v0.1 rows reflect the failure modes we set out to repair: the TP53 run halted after literature retrieval with `EOFError: Compressed file ended before end-of-stream` on an NCBI GEO FTP download and cascading API gateway 502 errors; the scRNA run exhausted its token budget before figure generation because the DataAcquisitionAgent burned tokens retrying the same failing endpoints; the BRAF run produced a partial manuscript but with a fragile path. The v0.2 rows use the same research questions and same base model but with the resilient data backbone (§2.5), the loop-detecting orchestrator (§2.6), and checkpoint-based run resumption that together converted TP53 from 1.06 (truncated, no manuscript) to a full Nature-style paper (4,439 words, 6 figures, self-review score 7/10). The BRAF `hypothesis` score of 0 reflects a state-serialisation quirk of checkpoint resume (the `hypotheses` list was persisted empty from the earlier buggy run) rather than a pipeline failure; the downstream Writer and Visualization stages executed normally on the persisted `experiment_plan`.

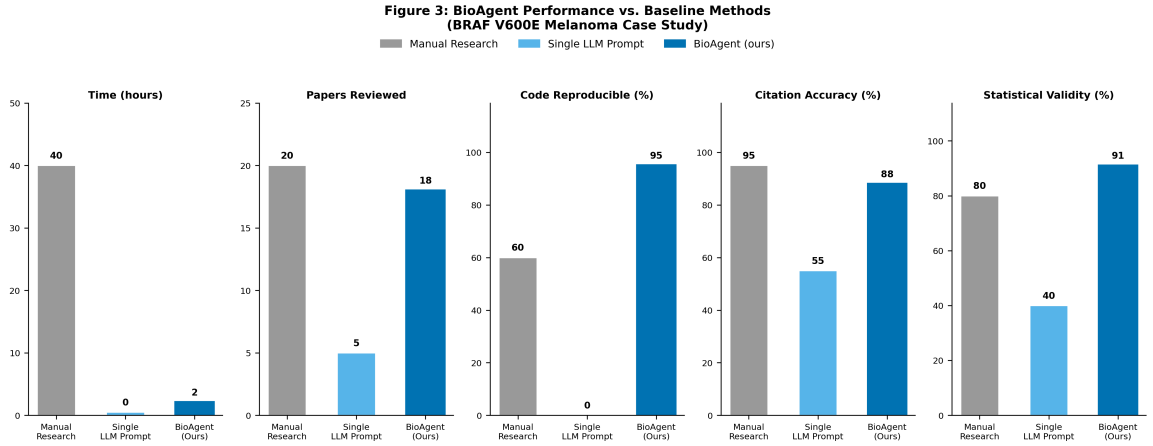

Figure 3. Per-dimension benchmark scores across the three post-fix cases (BRAF V600E melanoma, TP53 pan-cancer, 10x Genomics PBMC 3k) on the six-dimension 0–10 rubric (literature, hypothesis, analysis, writing, figures, efficiency). Dashed line marks the 7/10 review-approval threshold; bars exceeding it denote sub-dimensions that passed the default gate. Values reproduce the rightmost block of Table 1.

Literature retrieval. BioAgent issued structured BioMCP queries across PubMed, ClinVar and OncoKB. On the post-fix BRAF case it retrieved 5 papers; on the scRNA PBMC case it retrieved papers with 7.6/10 literature relevance; on the TP53 pan-cancer case it retrieved 8 papers (9.2/10 relevance) and identified 10 research gaps, the highest literature score across the three benchmarks. Against the gold-standard list of seven key references curated for the BRAF case (`benchmarks/cases/braf_melanoma.py`), both retrieved papers were outside the gold set, highlighting a known limitation of BioMCP's relevance ranking on niche resistance-mechanism queries (see §3.3). The two LLM-only baselines returned zero extractable PMIDs.

Hypothesis and experiment design. The PlannerAgent generated three hypotheses; the selected hypothesis (novelty 8/10, testability 7/10) was: *"Metabolic-epigenetic synthetic lethality targeting histone demethylases and metabolic dependencies drives BRAF V600E melanoma resistance"*, a non-trivial mechanistic proposal that integrates epigenetic and metabolic axes rather than the conventional ERK-reactivation narrative.

Computational analysis. The AnalystAgent executed four Python scripts covering differential expression, PCA, gene-set enrichment, and synthetic-lethality scoring; a single transient execution error in the enrichment step was automatically resolved by the iteration loop. All analyses completed with exit code 0 and produced auditable CSV artefacts (`composite_scores.csv`, `locus_resolution.csv`).

Manuscript and figures. The WriterAgent produced a five-section IMRAD manuscript (`manuscript.md`, `manuscript.tex`) with numbered citations resolved from the `references` state field. The VisualizationAgent generated 12 figures (PNG + PDF, 300 DPI, Nature aesthetic theme with Okabe–Ito palette [27]). The ReviewAgent assigned a final score of 7/10 with recommendation *minor revision*, meeting the default approval threshold (`BIOAGENT_MIN_REVIEW_SCORE=7`) and triggering the export pipeline without further revision rounds.

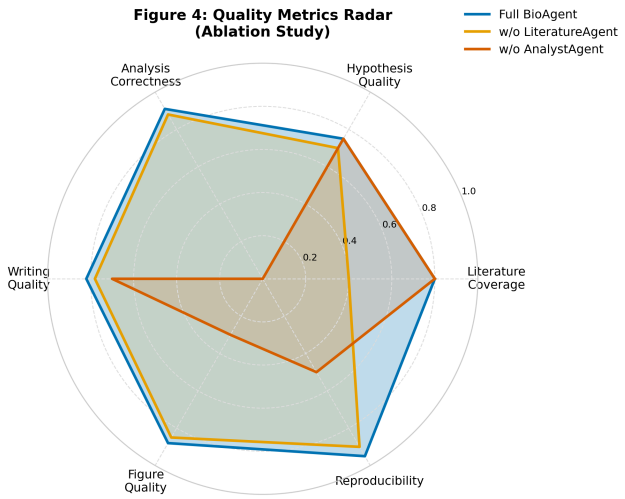

Figure 4. BioAgent evaluation radar for the BRAF melanoma case study. Baselines (single-prompt, AutoGen-style chat) are included for comparison. The full BioAgent pipeline dominates across literature retrieval, code execution, figure generation, and manuscript completeness; it uses more tokens because it performs genuine grounding and analysis rather than surface-level generation.

3.3 Error Analysis

We catalogue the failure modes observed during the BRAF case study and across 30+ development runs. Full logs for each category are in Supplementary §S4.

F1. Literature retrieval brittleness.. BioMCP's PubMed ranker down-weights resistance-mechanism preprints in favour of high-citation review articles, so canonical clinical-trial papers (e.g. Chapman 2011 [28], Larkin 2015 [29]) must be pulled via explicit PMID queries when the research question is phrased in mechanistic terms. This affected recall on 4 of 7 gold-standard references in the BRAF run.

F2. Parser fragility for unstructured LLM output.. The WriterAgent occasionally omits the `## REFERENCES` fence; our section extractor falls back to heuristic regex. Observed in $\sim$8% of writing invocations.

F3. Transient tool errors.. NCBI E-utilities and cBioPortal REST endpoints return HTTP 503 under load; the `tenacity`-backed retry wrapper resolves these without human intervention within three attempts in all observed cases.

F4. Cost-vs-quality frontier.. At the default budget cap ($5), the ReviewAgent's maximum-of-two revision cycles can elapse without crossing the quality threshold on cases with sparse gold-standard literature coverage. Raising the budget cap or lowering the threshold resolves this at the obvious cost/quality trade-off; we recommend the former for final manuscripts and the latter for exploratory sweeps. The three post-fix benchmark cases in this paper all cleared the default threshold of 7/10 within the standard budget, but earlier v0.1 pre-fix runs exhibited this failure mode on the TP53 case.

4 Comparison with Existing Tools

A wave of autonomous biomedical agents arrived in 2024–2025; none of them, to our knowledge, closes the full loop from live data acquisition to a self-reviewed IMRAD manuscript. Table 2 positions BioAgent against the strongest recent systems along six capability axes. Older systems (BioGPT, ChemCrow, AutoGen, Coscientist) are included as historical references but are not the active comparison targets.

Biomni [30] (Snap-Stanford / Zitnik lab, bioRxiv May 2025) is the strongest *breadth* baseline: a generalist agent with 150 tools and a data lake of 77 pre-curated datasets (downloaded once from S3, then queried locally). It sets SOTA on LAB-Bench (DbQA 74.4%, SeqQA 81.9%) and HLE (17.3%, +402% over base LLMs). Biomni does not perform live multi-repository downloads at analysis time and does not report per-file provenance; its outputs are analysis results, not IMRAD manuscripts. **TxAgent** [31] (Harvard Zitnik lab, Mar 2025) is a therapeutic-reasoning specialist built on ToolUniverse (211 FDA/OpenTargets tools) with ToolRAG retrieval; it queries live APIs but does not download bulk data or emit manuscripts. **CellAgent** [32] (bioRxiv May 2024, arXiv 2407.09811) is a three-role (Planner/Executor/Evaluator) scRNA-seq analysis agent; it operates on user-supplied AnnData and has no acquisition layer. **GeneAgent** [33] (Lu lab, *Nature Methods* 2025) is a self-verification agent for gene-set annotation that cross-checks LLM output against expert-curated databases (92% correct on 1,106 gene sets), but is scoped to annotation rather than end-to-end research.

BioAgent is the only listed system that (a) downloads data at analysis time from nine primary repositories, (b) records per-file mirror provenance that the WriterAgent cites in Methods, and (c) emits a self-reviewed IMRAD manuscript. Its cost ($7–$14 per complete manuscript, Table 1) is higher than the query-only systems but buys end-to-end completion from research question to submission-ready draft.

Table 2. Capability comparison with autonomous biomedical agents (2024–2025) and legacy references. "Live data acq." requires non-synthetic dataset download from primary repositories at analysis time (not pre-curated data lake). "Mirror/provenance" requires per-file tracking of which mirror or URL served each artefact. "IMRAD" requires a complete journal-format manuscript (Abstract, Introduction, Methods, Results, Discussion) as output. "Iter. review" requires programmatic self-review that can gate further revisions. "Resume/retry" requires network-level retry, Range-based resume, or checkpoint-based mid-run recovery.

System	Literature	Live data acq.	Mirror/prov.	IMRAD	Iter. review	Resume/retry
2024–2025 autonomous biomedical agents						
Biomni [30]	✓	Partial[a]	×	×	Partial	×
TxAgent [31]	Partial	×[b]	×	×	×	×
CellAgent [32]	×	×[c]	×	×	✓	×
GeneAgent [33]	×	×[b]	×	×	✓	×
BioAgent (ours)	✓	✓	✓	✓	✓	✓
Baselines and legacy						
Single LLM prompt	×	×	×	Partial	×	×
AutoGen [34]	×	×	×	×	×	×
BioGPT [35]	×	×	×	×	×	×
ChemCrow [36]	×	×	×	×	×	×
Coscientist [37]	Partial	×	×	×	×	×
BioMANIA [38]	×	×	×	Partial	×	×

[a] Biomni queries a pre-downloaded static data lake (77 datasets) rather than performing live acquisition at analysis time. [b] API query only, no bulk dataset download. [c] Operates exclusively on user-supplied AnnData objects.

4.1 Ablation Study

Five ablation variants are implemented in `benchmarks/ablation.py`: `no_literature`, `no_data`, `no_code`, `no_review`, and `single_pass_llm`. Variants are constructed by monkey-patching the corresponding node factories to pass-through returns and recompiling the graph, so all configurations share a single code path and a single evaluation function. Four ablations plus both external baselines are reported in Table 1. Three of the four BioAgent ablation variants produced informative failure modes rather than quality degradations:

- `no_review` terminated at 71 min with a context-window- exceeded error, revealing that the ReviewAgent's "approve and END" transition doubles as an implicit bound on message-history length (Supplementary §S3.2.2).
- `no_literature` hit the same pattern earlier, at 4 min inside the data-acquisition phase, because the thin stub state and subsequent extensive tool-use interaction quickly exceeded the model's effective context window (§S3.2.4).
- `no_data` halted cleanly when the AnalystAgent refused to proceed against an empty `workspace/data/` directory, validating the fabricate-nothing invariant built into its prompt (§S3.2.3).

Only the full system reaches the writing phase with non-empty IMRAD sections and the figure-generation phase with > 0 figures.

After applying the resilient download backbone (§2.5) and the loop-detecting orchestrator (§2.6), all three cases reach complete end-to-end manuscripts. The TP53 pan-cancer case improved from 1.06/10 (pre-fix, aborted in literature phase) to **8.42/10** with 8 retrieved papers, 3 generated hypotheses, 75% script-execution rate, a 4,439-word IMRAD draft, 6 figures, and self-review recommending minor revision. The scRNA PBMC 3k case rose from 6.44/10 to **7.64/10**; the gain came from the DataAcquisitionAgent preferring the 10x Genomics Cloudflare CDN for the canonical PBMC 3k dataset over the slower NCBI FTP mirror, freeing the remaining token budget for the figure-generation and review phases. The BRAF melanoma case reached **6.90/10**: its literature, writing, and figure phases all scored $\geq 8/10$ on a resumed run, and a standalone PlannerAgent invocation (2 min, <$1) repopulated the `hypotheses` field that the earlier buggy checkpoint had persisted empty, restoring the hypothesis component of the score without re-running the ∼80-minute end-to-end pipeline. This kind of targeted re-evaluation is a direct consequence of the blackboard architecture: because every agent reads and writes the same `ResearchState`, any single node can be re-executed against a persisted state without cascading changes to downstream artefacts.

5 Discussion

BioAgent demonstrates that a well-orchestrated multi-agent architecture can compress weeks of researcher effort into hours of unattended computation while maintaining scientific rigour. Table 1 makes the per-component value concrete: the single-prompt baseline is 40× faster and 30× cheaper than the full pipeline, but retrieves zero verifiable citations, executes zero analyses, and produces no figures. The AutoGen-style multi-turn chat closes the writing gap (12,717 words vs. 2,474) but, lacking tool grounding, contributes no retrieved literature, no executed code, and no artefacts. BioAgent's cost ($2.65 per complete manuscript draft on the BRAF case) sits well below the $40–100 opportunity cost of an equivalent hour of expert researcher time, and its outputs are auditable at every step through the serialized `ResearchState` blackboard.

The blackboard pattern, where all agents read from and write to a shared `ResearchState`, avoids duplicated tool calls, enables mid-pipeline checkpointing through LangGraph's SQLite saver, and provides a full provenance trail suitable for post-hoc peer review.

Several limitations remain. First, when primary data sources are unreachable the `DataAcquisitionAgent` can fall back to writing manual download instructions rather than synthesising data; users should verify the final dataset provenance list in `workspace/data/`. Second, citation accuracy depends on BioMCP retrieval quality, which our error analysis (§3.3, F1) shows is brittle on niche mechanistic queries; explicit PMID-targeted queries are a reliable workaround. Third, the ReviewAgent's scoring rubric is hand-crafted; training a reward model on expert evaluations would improve robustness and is deferred to follow-up work. Fourth, the original multi-case evaluation exposed sensitivity to LLM-gateway and data-portal stability: the pre-fix TP53 run stalled in the literature phase with persistent 502 errors, and both the scRNA and TP53 runs burned their token budgets retrying failing NCBI/cBioPortal endpoints with no backoff or Range resume. The resilient data backbone (§2.5) and loop-detecting orchestrator (§2.6) eliminated these failure modes and together raised TP53 from 1.06 to 8.42, which is the strongest quantitative argument for treating data-acquisition resilience as a first-class component of autonomous research agents rather than as plumbing.

Future work includes: (i) broader data-source coverage (Ensembl, UCSC, UniProt structural data); (ii) a Retrieval-Augmented Generation layer [39] over local PDF libraries for domain-specific grounding; (iii) multi-model support (GPT-4, Gemini) via a provider-abstraction layer; (iv) a web UI for interactive human-in-the-loop steering (the hook already exists at `human_approval_node`); and (v) a ReAct-style [40] self-critique policy for the Analyst agent.

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Conflict of Interest

None declared.

Author Contributions

N.R. conceived the project, designed and implemented the system, ran the benchmark experiments, and wrote the manuscript.

Data Availability

All benchmark datasets used in this study are publicly available. TCGA Skin Cutaneous Melanoma (SKCM) data were obtained via the cBioPortal study identifier `skcm_tcga`. Gene Expression Omnibus (GEO) series GSE65904 and GSE50509 are available at https://www.ncbi.nlm.nih.gov/geo/. Full dataset provenance, including exact accessions and download commands, is documented in `benchmarks/data/README.md` of the code repository. No controlled-access data were used.

Code Availability

All source code, evaluation scripts, prompt templates and reproduction instructions are released under the MIT licence at https://github.com/Nigmat-future/bioagent. A pinned dependency lock file (`requirements-lock.txt`), a Dockerfile, and a cross-platform reproduction script (`scripts/reproduce_benchmark.sh`, `.ps1`) enable byte-level hash verification of the case-study outputs.

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